# CHARACTERIZATION OF THE STABLE CONTRACTION OF A NODAL CURVE OF GENUS AT LEAST 2 WITH H^0(X, O_X) = K

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: existence by the composition of tail and bridge contractions, the full fibre analysis of the competitor, the rational-tail-to-point claim with both cases, the induction over tail contractions, and the Proj comparison closing the uniqueness proof, including the author's remark that the uniqueness of the isomorphism is omitted as a verification.

*Editorial note.* The label lemma-characterize-contraction-to-stable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Contractions of rational tails and rational bridges. Same chapter, Lemmas lemma-contracting-rational-tails and lemma-contracting-rational-bridges with Examples example-rational-tail and example-rational-bridge. Supplies the existence half: a morphism c with O_Y = c_*O_X, R^1c_*O_X = 0 and ample omega_Y, obtained as a composition of tail and bridge contractions.
- Stein factorization and fibres of c. More on Morphisms, Theorem more-morphisms-theorem-stein-factorization-Noetherian and Lemmas more-morphisms-lemma-check-h1-fibre-zero, more-morphisms-lemma-h1-fibre-zero-check-h0-kappa. Gives geometrically connected fibres and the identities H^1(X_y, O) = 0, H^0(X_y, O) = kappa(y) used to classify the fibres of the competing contraction.

*Source reference.* Original lemma 02OG: sources/stacks-chow/tags.txt, source lines 3515–3515. Prerequisite pointer invoked at proof line(s) 6721; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E3M: sources/stacks-curves/pinned-original-source.tex, source lines 6201–6201. Prerequisite pointer invoked at proof line(s) 6681; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E3H: sources/stacks-curves/pinned-original-source.tex, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 6680; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0BY5: sources/stacks-curves/pinned-original-source.tex, source lines 1277–1277. Prerequisite pointer invoked at proof line(s) 6708; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E38: sources/stacks-curves/pinned-original-source.tex, source lines 4794–4794. Prerequisite pointer invoked at proof line(s) 6713; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E7P: sources/stacks-curves/pinned-original-source.tex, source lines 6572–6572. Prerequisite pointer invoked at proof line(s) 6771; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E3Q: sources/stacks-curves/pinned-original-source.tex, source lines 6491–6491. Prerequisite pointer invoked at proof line(s) 6671; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E3L: sources/stacks-curves/pinned-original-source.tex, source lines 6135–6135. Prerequisite pointer invoked at proof line(s) 6670; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E3K: sources/stacks-curves/pinned-original-source.tex, source lines 6080–6080. Prerequisite pointer invoked at proof line(s) 6768; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E3N: sources/stacks-curves/pinned-original-source.tex, source lines 6367–6367. Prerequisite pointer invoked at proof line(s) 6774; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0E63: sources/stacks-curves/pinned-original-source.tex, source lines 5987–5987. Prerequisite pointer invoked at proof line(s) 6738; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0B62: sources/stacks-curves/pinned-original-source.tex, source lines 1354–1354. Prerequisite pointer invoked at proof line(s) 6770; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0E7F: sources/stacks-chow/tags.txt, source lines 17780–17780. Prerequisite pointer invoked at proof line(s) 6702; it names a standard

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0E7I: sources/stacks-chow/tags.txt, source lines 17783–17783. Prerequisite pointer invoked at proof line(s) 6704; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original theorem 03H0: sources/stacks-chow/tags.txt, source lines 4547–4547. Prerequisite pointer invoked at proof line(s) 6693; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0C6J: sources/stacks-chow/tags.txt, source lines 15299–15299. Prerequisite pointer invoked at proof line(s) 6784; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0344: sources/stacks-chow/tags.txt, source lines 4079–4079. Prerequisite pointer invoked at proof line(s) 6711; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 0BXX: sources/stacks-curves/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 6751; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $k$ be a field. Let $X$ be a proper scheme over $k$ of dimension 1 with $H^0(X, \mathcal{O}_X) = k$ having genus $g \geq 2$. Assume the singularities of $X$ are at-worst-nodal. There is a unique morphism (up to unique isomorphism)*

$$c : X \longrightarrow Y$$

*of schemes over $k$ having the following properties:*

1. *$Y$ is proper over $k$, $\dim(Y) = 1$, the singularities of $Y$ are at-worst-nodal,*
2. *$\mathcal{O}_Y = c_*\mathcal{O}_X$ and $R^1c_*\mathcal{O}_X = 0$, and*
3. *$\omega_Y$ is ample on $Y$.*

*Proof.* Existence: A morphism with all the properties listed exists by combining Lemmas 0E3L and 0E3Q as discussed in the introduction to this section. Moreover, we see that it can be written as a composition

$$X \to X_1 \to X_2 \ldots \to X_n \to X_{n+1} \to \ldots \to X_{n+n'}$$

where the first $n$ morphisms are contractions of rational tails and the last $n'$ morphisms are contractions of rational bridges. Note that property (2) holds for each contraction of a rational tail (Example 0E3H) and contraction of a rational bridge (Example 0E3M). It is easy to see that this property is inherited by compositions of morphisms.

Uniqueness: Let $c : X \to Y$ be a morphism satisfying conditions (1), (2), and (3). We will show that there is a unique isomorphism $X_{n+n'} \to Y$ compatible with the morphisms $X \to X_{n+n'}$ and $c$.

Before we start the proof we make some observations about $c$. We first observe that the fibres of $c$ are geometrically connected by More on Morphisms, Theorem 03H0. In particular $c$ is surjective. For a closed point $y \in Y$ the fibre $X_y$ satisfies

$$H^1(X_y, \mathcal{O}_{X_y}) = 0 \quad \text{and} \quad H^0(X_y, \mathcal{O}_{X_y}) = \kappa(y)$$

The first equality by More on Morphisms, Lemma 0E7F and the second by More on Morphisms, Lemma 0E7I. Thus either $X_y = x$ where $x$ is the unique point of $X$ mapping to $y$ and has the same residue field as $y$, or $X_y$ is a 1-dimensional proper scheme over $\kappa(y)$. Observe that in the second case $X_y$ is Cohen-Macaulay (Lemma 0BY5). However, since $X$ is reduced, we see that $X_y$ must be reduced at all of its generic points (details omitted), and hence $X_y$ is reduced by Properties, Lemma 0344. It follows that the singularities of $X_y$ are at-worst-nodal (Lemma 0E38). Note that the genus of $X_y$ is zero (see above). Finally, there are only a finite number of points $y$ where the fibre $X_y$ has dimension 1, say $\{y_1, \ldots, y_r\}$, and $c^{-1}(Y \setminus \{y_1, \ldots, y_r\})$ maps isomorphically to $Y \setminus \{y_1, \ldots, y_r\}$ by $c$. Some details omitted; hint: outside of $\{y_1, \ldots, y_r\}$ the morphism $c$ is finite, see Cohomology of Schemes, Lemma 02OG.

Let $C \subset X$ be a rational tail. We claim that $c$ maps $C$ to a point. Assume that this is not the case to get a contradiction. Then the image of $C$ is an irreducible component $D \subset Y$. Recall that $H^0(C, \mathcal{O}_C) = k'$ is a finite separable extension of $k$ and that $C$ has a $k'$-rational point $x$ which is also the unique intersection of $C$ with the "rest" of $X$. We conclude from the general discussion above that $C \setminus \{x\} \subset c^{-1}(Y \setminus \{y_1, \ldots, y_r\})$ maps isomorphically to an open $V$ of $D$. Let $y = c(x) \in D$. Observe that $y$ is the only point of $D$ meeting the "rest" of $Y$. If $y \notin \{y_1, \ldots, y_r\}$, then $C \cong D$ and it is clear that $D$ is a rational tail of $Y$ which is a contradiction with the ampleness of $\omega_Y$ (Lemma 0E63). Thus $y \in \{y_1, \ldots, y_r\}$ and $\dim(X_y) = 1$. Then $x \in X_y \cap C$ and $x$ is a smooth point of $X_y$ and $C$ (Lemma 0E38). If $y \in D$ is a singular point of $D$, then $y$ is a node and then $Y = D$ (because there cannot be another component of $Y$ passing through $y$ by Lemma 0E38). Then $X = X_y \cup C$ which means $g = 0$ because it is equal to the genus of $X_y$ by the discussion in Example 0E3H; a contradiction. If $y \in D$ is a smooth point of $D$, then $C \to D$ is an isomorphism (because the nonsingular projective model is unique and $C$ and $D$ are birational, see Section 0BXX). Then $D$ is a rational tail of $Y$ which is a contradiction with ampleness of $\omega_Y$.

Assume $n \geq 1$. If $C \subset X$ is the rational tail contracted by $X \to X_1$, then we see that $C$ is mapped to a point of $Y$ by the previous paragraph. Hence $c : X \to Y$ factors through $X \to X_1$ (because $X$ is the pushout of $C$ and $X_1$, see discussion in Example 0E3H). After replacing $X$ by $X_1$ we have decreased $n$. By induction we may assume $n = 0$, i.e., $X$ does not have a rational tail.

Assume $n = 0$, i.e., $X$ does not have any rational tails. Then $\omega_X^{\otimes 2}$ and $\omega_X^{\otimes 3}$ are globally generated by Lemma 0E3K. It follows that $H^1(X, \omega_X^{\otimes 3}) = 0$ by Lemma 0B62. By Lemma 0E7P applied with $m = 3$ we find that $c^*\omega_Y \cong \omega_X$. We also have that $\omega_X = (X \to X_{n'})^*\omega_{X_{n'}}$ by Lemma 0E3N and induction. Applying the projection formula for both $c$ and $X \to X_{n'}$ we conclude that

$$\Gamma(X_{n'}, \omega_{X_{n'}}^{\otimes m}) = \Gamma(X, \omega_X^{\otimes m}) = \Gamma(Y, \omega_Y^{\otimes m})$$

for all $m$. Since $X_{n'}$ and $Y$ are the Proj of the direct sum of these by Morphisms, Lemma 0C6J we conclude that there is a canonical isomorphism $X_{n'} = Y$ as desired. We omit the verification that this is the unique isomorphism making the diagram commute. $\square$